

Project Executable Files

Date	1 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	yes
3	requirements.txt / package.json / dependency file	yes
4	Database schema / seed files (if applicable)	yes
5	Environment configuration file (.env.example or similar)	NA – no API keys or secrets used
6	Deployed application URL (if hosted)	yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	yes
8	Dockerfile / Containerization config (if applicable)	yes
9	Test files and test results	yes
10	Demo video or walkthrough (if required)	yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```

Rising_Waters_Project
|
+-- app.py
+-- flood dataset.xlsx
+-- floods.save
+-- transform.save
+-- main.ipynb
|
+-- static/
|   +-- main.css
|   +-- main.js
|
+-- templates/
    +-- home.html
    +-- index.html
    +-- chance.html
    +-- no_chance.html
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://smart-bridge-rising-waters.onrender.com
Login Credentials (Demo)	NA
Platform / Hosting Provider	Render
Repository Link	https://github.com/Gayatri344/smart--bridge_Rising-Water
Demo Video Link	https://drive.google.com/file/d/1pG4X3EnbLGHCGfTuvw1CpYu_kS9g3GzV/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

- 1 Download or clone the project folder (Rising_Waters_Project).
- 2 Open the project folder in Visual Studio Code, PyCharm, or any Python IDE.
- 3 Install the required libraries: Flask, NumPy, Pandas, Scikit-learn, Joblib (`pip install flask numpy pandas scikit-learn joblib`).
- 4 Verify that all project files are present, including `app.py`, `floods.save`, `transform.save`, the `static/` folder, and the `templates/` folder.
- 5 Start the Flask application by running: `python app.py`
- 6 Open a web browser and navigate to: `http://127.0.0.1:5000`
- 7 From the Home page, click "Launch Predictive Tool" to open the Prediction Input page.
- 8 Enter the required weather parameters: Cloud Cover (%), Annual Rainfall (mm), Jan–Feb Rainfall (mm), Mar–May Rainfall (mm), and Jun–Sep Rainfall (mm).
- 9 Click "Execute Prediction Model" to generate the flood risk result.
- 10 View the result on the Critical Flood Risk page or the Environmental Status Secure page.

Online Access

Not yet deployed online. Once IBM Cloud deployment is complete, the public URL will be added here.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	The saved model (<code>floods.save</code>) was trained on the wrong target column (sub instead of flood), so the deployed app currently routes every prediction to the flood-risk warning page regardless of input.	Retrain using the flood column as the target, fix the scaler fit on the test set, and re-save <code>floods.save</code> / <code>transform.save</code> before demo or grading.
2	The model saved as <code>floods.save</code> is a scikit-learn <code>GradientBoostingClassifier</code> , not the XGBoost model implied by <code>app.py</code> comments and earlier project documentation.	Either train a genuine <code>xgb.XGBClassifier</code> , or update comments/docs to accurately describe the Gradient Boosting model in use.
3	The training dataset is small and imbalanced (115 records: 99 non-flood vs. 16 flood).	Collect additional historical records, or apply class weighting / resampling when retraining.
4	The application is not yet deployed to a public URL.	Deploy to IBM Cloud as planned in the Technology Stack document.